

REMARKS ON SPLITTING CONICS OF SMOOTH PLANE SEXTICS

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ABSTRACT.

We work over the complex number field $\mathbb{C}$.

Let B be a smooth plane sextic. A smooth conic C is said to be a splitting conic of B if the local intersection multiplicity of B and C at every point of $B \cap C$ is even. A splitting-conic configuration is a pair (B, Γ) of a smooth plane sextic B and a non-ordered collection $\Gamma = \{C_1, \dots, C_n\}$ of splitting conics of B . The number n of the conics in Γ is the size of the configuration. A splitting-conic configuration (B, Γ) of size n is said to be of general singularity type if each member C_i of Γ intersects B at distinct six points and no three curves in $B, C_1, \dots, C_n$ have a common point, or equivalently, the singulatieis of the union of $B, C_1, \dots, C_n$ consists of $6n$ tacnodes and $n(n-1)/2$ ordinary nodes.

We investigate splitting configurations of smooth plane sextics. Our main results are the following.

- (A) We exhibit a splitting-conic configuration of size 120, which is associated with the del Pezzo surface of degree 1 and the root lattice of type E_8 .
- (B) We give the complete classification of splitting-conic configurations of size ≤ 6 up to deformation equivalence.
- (C) We construct examples of splitting-conic configurations of large sizes by means of the Leech lattice. The maximum size we have found is 4455.

As the phrase “splitting” suggests, the main idea of our investigation is to consider the double covering

$$\pi_B: X_B \rightarrow \mathbb{P}^2$$

branched along the smooth plane sextic B . The double plane X_B is a K3 surface, and the inverse image $\pi_B^{-1}(C)$ of a splitting conic C of B splits into two (-2) -curves C^+ and C^- , which we call the lifts of C . For distinct splitting conics C_i and C_j of B , we have

$$\langle C_i^+, C_j^+ \rangle \in \{0, 1, 2, 3, 4\}, \quad \langle C_i^+, C_j^+ \rangle + \langle C_i^-, C_j^+ \rangle = 4.$$

To describe the intersection patterns of the lifts of splitting conics, we introduce an invariant $\gamma(B, \Gamma)$. Let $\mathcal{G}_n$ be the set of weighted complete graphs with the set of vertices $\{1, \dots, n\}$, and each edge ij ($i \neq j$) is weighted by an integer $w_{ij} \in$

$\{0, 1, 2, 3, 4\}$. The group

$$G_n = \mathfrak{S}_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$$

acts on $\mathcal{G}_n$, where the symmetric group $\mathfrak{S}_n$ acts on $\mathcal{G}_n$ via permutations of labeling of the vertices, and the i -th factor $\mathbb{Z}/2\mathbb{Z}$ of $(\mathbb{Z}/2\mathbb{Z})^n$ acts on $\mathcal{G}_n$ by changing the weights w_{ij} of the edges containing the vertex i to $4 - w_{ij}$.

Definition 0.1. Let (B, Γ) be a splitting-conic configuration of size n . We label the members of Γ as $C_1, \dots, C_n$, and choose a lift C_i^+ of each member C_i . Then we obtain a weighted graph $g(B, \Gamma) \in \mathcal{G}_n$ by putting the weight w_{ij} to be $\langle C_i^+, C_j^+ \rangle$. Let $\gamma(B, \Gamma) \in \mathcal{G}_n/G_n$ be the G_n -orbit of the graph $g(B, \Gamma)$. Then it is obvious that $\gamma(B, \Gamma)$ does not depend on the choice of the labelling of Γ and the choices of the lifts C_i^+ .

(A) From the theory of del Pezzo surfaces [?], we obtain the following simple observation:

Proposition 0.2. *For $\nu = 0, 2, 4, 6$, let $f_\nu(x_0, x_1)$ be a general homogeneous polynomial of degree ν . Then the smooth plane sextic B_ν defined by the equation*

$$(0.1) \quad f_6(x_0, x_1) + f_4(x_0, x_1)x_2^2 + f_2(x_0, x_1)x_2^4 + f_0(x_0, x_1)x_2^6 = 0$$

possesses exactly 120 splitting conics. Let Δ_{120} be the union of these conics. Then (B_ν, Δ_{120}) is of general singularity type.

The invariant $\gamma(B_\nu, \Delta_{120})$ is calculated by the following. Let $\mathbb{E}_8$ denote the positive definite root lattice of type E_8 with the intersection form $(\cdot, \cdot)_E$, and let $(\mathbb{E}_8)_2$ denote the set of vectors r of norm $(r, r)_E = 2$.

Proposition 0.3. *Let $\tilde{\Delta}_{120}$ be the set of lifts of splitting conics of B_ν , on which the switching $C^+ \leftrightarrow C^-$ acts. Then there exists a bijection*

$$\rho: \tilde{\Delta}_{120} \xrightarrow{\sim} (\mathbb{E}_8)_2$$

such that

- (i) *the switching on $\tilde{\Delta}_{120}$ corresponds to the involution $r \leftrightarrow -r$ on $(\mathbb{E}_8)_2$, and*
- (ii) *for $\tilde{C}, \tilde{C}' \in \tilde{\Delta}_{120}$, we have*

$$\langle \tilde{C}, \tilde{C}' \rangle = -2(\rho(\tilde{C}), \rho(\tilde{C}'))_E + 2.$$

We construct examples of Zariski multiples from the configuration (B_ν, Δ_{120}) .

Definition 0.4. Let D and D' be reduced, possibly reducible plane curves. We say that D and D' are of the same combinatorial type if there exist tubular neighborhoods T of D and T' of D' in $\mathbb{P}^2$ such that (T, D) and (T', D') are homeomorphic, whereas we say that D and D' are of the same topological type if $(\mathbb{P}^2, D)$ and $(\mathbb{P}^2, D')$ are homeomorphic.

We say that a collection of reduced plane curves $D_1, \dots, D_N$ form a Zariski N -uple if they are of the same configuration type, but any two curves D_i, D_j ($i \neq j$) are of different topological types.

For a subset Δ of Δ_{120} , let D_Δ denote the union of B_ι and the conics in Δ . Note that the combinatorial type of D_Δ depends only on the size of Δ .

We put

$$\mathcal{S}_n := \left\{ \{r_1, \dots, r_n\} \subset (\mathbb{E}_8)_2 \mid \begin{array}{l} r_1, \dots, r_n \in \mathbb{E}_8 \text{ are distinct vectors of norm } 2 \\ \text{that span } \mathbb{E}_8 \otimes \mathbb{Q} \text{ over } \mathbb{Q} \end{array} \right\},$$

on which the Weyl group $W(\mathbb{E}_8)$ acts.

Theorem 0.5. *The set of plane curves D_Δ , where Δ runs through the subsets of Δ_{120} of size $|\Delta| = n$, are divided into at least S_n distinct topological types, where s_n is the number of $W(\mathbb{E}_8)$ -orbits in $\mathcal{S}_n$.*

See Remark ?? for a rough estimate of the number s_n . Thus we obtain examples of Zariski N -ples for large N by choosing various subsets Δ of Δ_{120} .

See Section ?? for the proof of this proposition and further geometric properties of this configuration of 120 conics.

1. ENUMERATION OF SPLITTING CONICS

Let B be a smooth plane sextic, let $\pi_B: X_B \rightarrow \mathbb{P}^2$ be the double cover branched along B , and let $j: X_B \rightarrow X_B$ be the involution with respect to the double cover $\pi_B: X_B \rightarrow \mathbb{P}^2$. We denote by $h \in \text{Pic}(X_B)$ the class of the pull-back of a line of $\mathbb{P}^2$. Then h is ample. The involution j fixes h and acts as multiplication by -1 on the orthogonal complement $h^\perp$ in $\text{Pic}(X_B)$. The set $\mathcal{R}_d$ of classes of smooth rational curves C of degree $\langle C, h \rangle = d$ on X_B is enumerated inductively on d as follows:

$$\mathcal{R}_d := \left\{ r \in \text{Pic}(X_B) \mid \begin{array}{l} \langle r, r \rangle = -2, \langle r, h \rangle = d, \\ \text{for all } d' < d \text{ and for all } r' \in \mathcal{R}_{d'}, \text{ we have } \langle r, r' \rangle \geq 0 \end{array} \right\}.$$

Then the set of classes of splitting conics of B is equal to the set of j -orbits in $\mathcal{R}_2$.

2. SPLITTING CONIC CONFIGURATION (B_ι, Δ_{120})

2.1. Proof of Proposition 0.2. Throughout this subsection, let B denote a smooth plane sextic defined by an equation of the form (0.1). Let $\iota: \mathbb{P}^2 \rightarrow \mathbb{P}^2$ be the involution given by

$$[x_0 : x_1 : x_2] \mapsto [x_0 : x_1 : -x_2].$$

Then B is ι -invariant. We show that the Picard lattice $\text{Pic}(X_B)$ of the $K3$ surface X_B is of rank 9 and admits a basis such that

- (a) the Gram matrix of $\text{Pic}(X_B)$ is the diagonal matrix

$$\text{diag}(2, -2, \dots, -2),$$

(b) the class $h \in \text{Pic}(X_B)$ of the pull-back of a line of $\mathbb{P}^2$ is written as

$$h = (3, -1, \dots, -1).$$

We denote the space of all plane sextics by

$$\mathcal{P}_6 := \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6))),$$

and let $\mathcal{S} \subset \mathcal{P}_6$ be the Zariski open subset of smooth plane sextics. Then the period map gives an open immersion of $\mathcal{S}/\text{PGL}(3, \mathbb{C})$ into a 19-dimensional period domain. The space $\mathcal{I}_\iota$ of ι -invariant sextics is a 15-dimensional subspace of $\mathcal{P}_6$. Since the centralizer of ι in $\text{PGL}(3, \mathbb{C})$ is of dimension 4, the image of

$$\mathcal{I}_\iota \cap \mathcal{S} \hookrightarrow \mathcal{S} \rightarrow \mathcal{S}/\text{PGL}(3, \mathbb{C})$$

is $15 - 4 = 11$. (Here we use the fact that a smooth plane sextic is of genus 10 and hence its automorphism group is finite.) Since B is a general point of $\mathcal{I}_\iota \cap \mathcal{S}$, the rank of $\text{Pic}(X_B)$ is $20 - 11 = 9$. Let $\phi: \mathbb{P}^2 \rightarrow Q$ be the quotient morphism by the involution ι , where $Q \subset \mathbb{P}^3$ is the quadric surface of rank 3 defined by $z_0 z_2 - z_1^2 = 0$, and ϕ is given by

$$[x_0 : x_1 : x_2] \mapsto [z_0 : z_1 : z_2 : z_3] = [x_0^2 : x_0 x_1 : x_1^2 : x_2^2].$$

The fixed point $[0 : 0 : 1]$ of ι is mapped to the vertex A of Q and the fixed line $\{x_2 = 0\}$ is mapped to the plane section $Q \cap H_0$ of Q , where $H_0 = \{z_3 = 0\}$. The quotient morphism ϕ induces a double covering $B \rightarrow B_Q$ of a $(2, 3)$ -complete intersection $B_Q \subset Q$. Let $\varpi_B: Y_B \rightarrow Q$ be the double covering branched along B_Q and at the vertex A . Then Y_B is a del Pezzo surface of degree 1, and the morphism $\varpi_B: Y_B \rightarrow Q$ is the bi-anticanonical model of Y_B . We denote by α_W the anti-canonical class of Y_B . The morphism ϕ lifts to a double covering $\varphi: X_B \rightarrow Y_B$. Thus we obtain the following commutative diagram:

$$\begin{array}{ccc} X_B & \xrightarrow{\varphi} & Y_B \\ \pi_B \downarrow & & \downarrow \varpi_B \\ \mathbb{P}^2 & \xrightarrow[\phi]{} & Q. \end{array}$$

The branch locus of the double covering $\varphi: X_B \rightarrow Y_B$ is the pull-back $\varpi_B^{-1}(Q \cap H_0)$, which is a member of the bi-anticanonical system $|2\alpha_W|$. Since the pull-back by ϕ of a plane section $Q \cap H$ of Q is a conic on $\mathbb{P}^2$, we have $2h = \varphi^*(2\alpha_W)$. Since $\text{Pic}(X_B)$ is torsion free, we obtain

$$h = \varphi^* \alpha_W.$$

The del Pezzo surface Y_B is obtained by blowing up a projective plane $\mathbf{P}^2$ at eight distinct points in general position. Hence $\text{Pic}(Y_B)$ is isomorphic to $I_{1,8}$, and has a basis $l, e_1, \dots, e_8$, where l is the class of the pull-back of a line of $\mathbf{P}^2$ and $e_1, \dots, e_8$ are the classes of exceptional divisors. We have

$$\alpha_W = 3l - e_1 - \dots - e_8.$$

$\dim_{\mathbb{F}_2} M/I_{1,8}(2)$	number	type of $R(h^\perp)$
0	1	0
1	135	$8A_1$
2	1575	$2D_4$
3	2025	D_8
4	270	E_8

TABLE 2.1. Overlattices M of $I_{1,8}(2)$

Now we have

$$I_{1,8}(2) \cong \varphi^* \text{Pic}(Y_B) \subset \text{Pic}(X_B).$$

Since $\text{Pic}(X_B)$ is of rank 9, $\text{Pic}(X_B)$ is an overlattice of $I_{1,8}(2)$. The discriminant form of $I_{1,8}(2)$ has 135 non-zero isotropic vectors. By Nikulin's theory of discriminant forms [?], we see that $I_{1,8}(2)$ has $1 + 135 + 1575 + 2025 + 270$ even overlattices as in Table 2.1. Note that

$$h = \varphi^*(3l - e_1 - \cdots - e_8)$$

is ample, and hence

$$R(h^\perp) := \{ r \in \text{Pic}(X_B) \mid \langle r, r \rangle = -2, \langle r, h \rangle = 0 \}$$

should be empty. On the other hand, as is shown in Table 2.1, if the extension index $\text{Pic}(X_B)/I_{1,8}(2)$ is > 1 , then $R(h^\perp)$ would have an element. Therefore the overlattice $\text{Pic}(X_B)$ is in fact equal to $\varphi^* \text{Pic}(Y_B)$ and hence has a basis $\varphi^*l, \varphi^*e_1, \dots, \varphi^*e_8$ with the desired properties (a), (b).

Then we see that X_B has exactly 120 splitting conics by the method given in Section ???. Indeed, there are no vectors r in $\text{Pic}(X_B)$ with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 1$, and there exists exactly 240 vectors r with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 2$. Therefore these 240 vectors are the classes of smooth rational curves. Moreover the involution j of the double covering $X_B \rightarrow \mathbb{P}^2$ acts on these 240 vectors without fixed points, and hence the 120 orbits under the involution j are in one-to-one correspondence with splitting conics.

In fact, these 240 vectors r are the pull-back of the 240 lines of the del Pezzo surface Y_B . Note that the mapping $B \mapsto B_Q$ gives an isomorphism between the space the projective space $\mathcal{I}_\iota$ of ι -invariant sextics and the linear system $|\mathcal{O}_Q(3)|$. Hence Y_B is a general member of the family $\mathcal{U}$ of bi-anticanonical models of del Pezzo surfaces of degree 1 studied in [?]. By [?, Proposition 4.4], the union of the 240 lines of Y_B has only ordinary double points, which implies that the 120 splitting conics of B are in fact six-tangential conics.

Remark 2.1. In fact, the automorphism group of the $K3$ surface X_B is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and the 240 curves obtained above are all the smooth rational curves on X_B .

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